

Stabilization of a Ball on a Stewart Platform:

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Mathematical Methods for Engineers - Final Project

1. Introduction

The Stewart Platform is a parallel manipulator consisting of a rigid top plate (which can be treated as its end effector) fixed to a base by six independently controlled linear actuators (legs). Since each leg is mounted to the end effector and has its own kinematic chain from base to end, it falls under the category of parallel manipulators. As opposed to serial link robots, where each joint can be described completely in the basis of the previous, parallel manipulators have joints in the common world frame that have a combined effect on the end effector (the rigid plate). The mechanism was proposed by D. Stewart in 1965 as a mechanism to simulate a plane's maneuverability. However, it has circulated modern day controls system engineering education as a classic project/example for teaching parallel manipulators in general, and it makes for a uniquely apt application of linear algebra and dynamical systems (Stewart, 1965).

The objective of this project is to design a feedback control law that commands the platform's tilt angles in real time so that for a given initial position and velocity of the ball, it can be driven to the target position (to simplify error calculations, the target position will be the center of the plate at position $= (0,0)$). On an uncontrolled surface, we will show that a ball is unstable and will roll away indefinitely the minute a small perturbation applies to such a surface.

The mathematical tools from Chapters 1 through 3 of the course notes provide the Linear Algebra, Laplace Transformations, and Dynamical Systems needed to analyze the system as designed, derive plant models, and design a PD controller in the frequency domain. First, the platform's orientation will be solved as a linear system to compute the desired leg

lengths for any given configuration. This system will have a transfer function that can be used as a plant model when developing the PD feedback controller. To analyze the stability and phase of the controller, we use the dynamical system analysis of a closed-loop controller. These analyses and controllers can then be implemented in MATLAB to simulate a ball-on-plate system.

2. Platform Geometry and Kinematics

Two coordinate frames need to be introduced to be able to describe this system: the world frame (defined as the center of the base plate, with positive-z pointing up) and the body frame (defined as the center of the top plate, with positive-z pointing up). When the platform is in its neutral position, the body frame is aligned with the world frame but is offset by an initial height h_0 .

The configuration space (i.e. the orientation) of the top plate is described by two variables: θ describes the rotation about the y-axis (also known as pitch) and ϕ describes the rotation about the x-axis (also known as roll). These values can be used to describe two rotation matrices, which when multiplied together show the rotation matrix for the system overall.

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

The combined rotation of the top plate is $R = R_x(\phi) R_y(\theta)$. Since matrix multiplication is generally non-commutative (as established in Theorem 1.3.1), the order

of application matters. Throughout this analysis, pitch is applied first, then roll. For small tilt angles ($|\theta|, |\varphi| \leq 20^\circ$), the product simplifies to the first-order approximation:

$$R(\theta, \varphi) \approx \begin{bmatrix} 1 & 0 & \theta \\ 0 & 1 & -\varphi \\ -\theta & \varphi & 1 \end{bmatrix} \quad (\text{Eq.2.1})$$

The Stewart platform can have many designs, the one chosen for this one is often called a hexapod arrangement. The arrangement has six legs, each connecting a base anchor point b_i to a top anchor point t_i . The base anchors are fixed in the world frame; the top anchors are defined in the body frame and rotate with the plate. The hexapod arrangement, as seen in Figure 1, uses an A-frame: the six legs are grouped into three pairs located at 0° , 120° , and 240° around the platform. Within each pair, the two base anchors are spread wide apart on a large-radius circle while the two top anchors sit close together on a small-radius circle. This geometry causes each pair of legs to form a visible triangular A-frame: wide at the base and narrow at the crown, exactly as observed in physical Stewart platforms. This geometry gives the platform an $h_0 = 0.9 \text{ m}$.

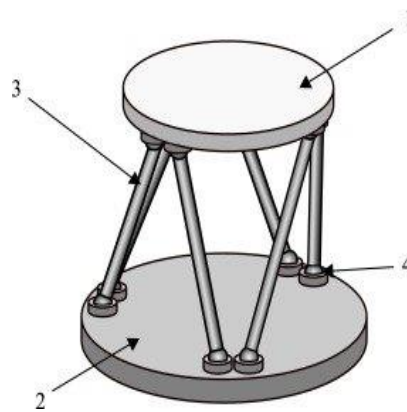


Figure 1. Reference Stewart platfor: three pairs of legs spread wide at the base (label 2) and converge toward the top plate (label 1)

For a given rotation R and platform center position $p = [0, 0, H_0]^T$, the world-frame position of top anchor i is:

$$t_i = R t_i + p \quad (\text{Eq. 2.2})$$

where t_i is the body-frame position of the anchor. The length of leg i is then simply the Euclidean norm of the difference vector:

$$\ell_i = \| t_i - b_i \| \quad (\text{Eq. 2.3})$$

Computing the six leg lengths from a known orientation creates an inverse kinematics problem, which is therefore straightforward using equation (2.3) directly. To study how small changes in tilt angles can be achieved from changes in leg lengths, we linearize about the neutral configuration. Let $\Delta\theta$ and $\Delta\phi$ be small perturbations in pitch and roll. The resulting perturbation in the position of top anchor i is:

$$\Delta t_i \approx (\Delta R) t_i \quad (\text{Eq. 2.4})$$

where ΔR is a small perturbation of the rotation matrix. The corresponding change in leg length i is the projection of this position perturbation onto the unit vector of leg i :

$$\Delta \ell_i = \hat{e}_i^T \Delta t_i \quad (\text{Eq. 2.5})$$

Stacking the six equations gives the Jacobian for the linear system:

$$J \Delta u = \Delta \ell \quad (\text{Eq. 2.6})$$

where $\Delta u = [\Delta\theta, \Delta\phi]^T$ is the vector of configuration space moving and $\Delta \ell = [\Delta \ell_1, \dots, \Delta \ell_6]^T$ is the vector of the actuation space moving, the leg length. J is defined as the 6×2 kinematic Jacobian to go from configuration space to actuation space. This is a

linear system of the form $Ax = b$, solvable by Gauss elimination (Section 1.4 and 1.5). The rank of J must equal to 2 for the platform to be able to achieve any desired configuration (since there are 2 variables θ and ϕ). A rank-deficient Jacobian corresponds to a singular configuration in which the platform loses a degree of freedom.

3. Ball Dynamics and State-Space Model

We now need to model a ping-pong ball of mass m and radius r . For simplicity, we will assume the ball rolls with no slip. The equations of motion for a solid sphere rolling without slipping on a tilted plate, linearized about the horizontal equilibrium under the small-angle assumption, are well established in the ball-and-plate control literature and referenced below [Awawdeh et al., 2024]

$$\ddot{x} = (5/7) g \theta \quad (3.1)$$

$$\ddot{y} = -(5/7) g \phi \quad (3.2)$$

Note that the negative sign in equation (3.2) is a consequence of the sign convention: a positive roll angle ϕ tilts the plate so that the ball accelerates in the negative y direction.

Equations (3.1) and (3.2) are second-order ordinary differential equations relating the configuration commands (θ, ϕ) —which the controller will provide—to the ball's acceleration. It is ideal that the x and y dynamics are completely decoupled: pitch θ controls only x , and roll ϕ controls only y . This allows the two-dimensional problem to be analyzed as two independent one-dimensional problems [Awatar et al., 2002].

3.2 State-Space Formulation

Each second-order equation is reduced to a system of first-order equations by introducing new variables for the velocity. We will complete this for equation (3.1), and since they are independent it is trivial to do the same for equation (3.2). Define the state vector for the x-axis as $z = [x_1, x_2]^T$ where $x_1 = x$ and $x_2 = \dot{x}$. Then from equation (3.1), we get:

$$\dot{x}_1 = x_2 \quad (3.3)$$

$$\dot{x}_2 = (5/7)g \theta \quad (3.4)$$

In matrix notation, using the formulation $\dot{z} = Az + Bu$:

$$\dot{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} = A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + B \begin{bmatrix} \theta \\ \varphi \end{bmatrix} \quad (3.5)$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 5g/7 & 0 \\ 0 & 0 \\ 0 & -5g/7 \end{bmatrix}$$

We now apply the eigenvalue analysis to the open-loop system matrix A. The characteristic polynomial is $\det(A - \lambda I) = \lambda^4$, whose only root is $\lambda = 0$ with algebraic multiplicity 4. By Theorem 3.4., since the matrix A has a zero eigenvalue, the equilibrium at the origin is neither an attractor nor a repeller—it is unstable in the sense that trajectories do not return to it.

This matches our expectations in open loop: the ball sitting at the center of a flat plate is in unstable equilibrium, any small perturbation will further away without bound until it falls off the platform. This confirms the necessity of active feedback control.

4. Controller Design via Laplace Transforms

4.1 Transfer Function of the Plant

We apply the Laplace transform to equation (3.1) to convert the ball dynamics into the frequency domain. Using Theorem 2.5.1 (Laplace of Derivatives), with zero initial conditions:

$$L\{\ddot{x}\} = s^2 X(s) - sx(0) - \dot{x}(0) = s^2 X(s) \quad (4.1)$$

$$L\{(5g/7)\theta\} = (5g/7) \Theta(s) \quad (4.2)$$

Combining these and using the linearity property of the Laplace transform:

$$s^2 X(s) = (5g/7) \Theta(s) \quad (4.3)$$

Solving for the transfer function $G(s) = X(s)/\Theta(s)$:

$$G(s) = X(s)/\Theta(s) = (5g/7) / s^2 \quad (4.4)$$

The double integrator is the canonical form for systems where an input (plate configuration angles) produces an acceleration of the output (ball position), and it requires a feedback controller to be stabilized.

4.2 PD Controller Design

A Proportional-Derivative (PD) controller is chosen, with transfer function:

$$C(s) = K_p + K_d s \quad (4.5)$$

The closed-loop transfer function, with unity negative feedback, is:

$$T(s) = C(s)G(s) / [1 + C(s)G(s)] \quad (4.6)$$

Substituting (4.4) and (4.5):

$$T(s) = [(5g/7)(K_d s + K_p)] / [s^2 + (5g/7)K_d s + (5g/7)K_p] \quad (4.7)$$

We match the denominator of (4.7) to the standard second-order form $s^2 + 2\zeta\omega_n s + \omega_n^2$ by equating coefficients:

$$\omega_n^2 = (5g/7) K_p \quad \Rightarrow \quad K_p = 7\omega_n^2 / (5g) \quad (4.8)$$

$$2\zeta\omega_n = (5g/7) K_d \quad \Rightarrow \quad K_d = 14\zeta\omega_n / (5g) \quad (4.9)$$

where ω_n is the desired natural frequency in radians per second and ζ is the desired damping ratio. These two parameters are the designer's tuning handles. A larger ω_n produces faster steady state convergence; a larger ζ reduces overshoot. The choice $\zeta = 1/\sqrt{2} \approx 0.707$ is often selected as a good compromise [Harib, et al., 2003]

Now, we can bring the response back in the time domain with inverse Laplace transforms. For a unit step input $X_{ref}(s) = 1/s$ and the closed-loop transfer function of (4.7), the output transform is found with a series of partial fractions:

$$X(s) = T(s) / s = \omega_n^2 / [s(s^2 + 2\zeta\omega_n s + \omega_n^2)]$$

$$X(s) = 1/s - (s + 2\zeta\omega_n) / (s^2 + 2\zeta\omega_n s + \omega_n^2)$$

$$x(t) = 1 - e^{(-\zeta\omega_n t)} [\cos(\omega_d t) + (\zeta/\sqrt{1-\zeta^2}) \sin(\omega_d t)] \quad (4.10)$$

where $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ is the damped natural frequency. Equation (4.10) exhibits the characteristic behavior of a second-order underdamped system: exponentially decaying oscillations about the setpoint.

5. Closed-Loop Stability Analysis

Having designed the gains K_p and K_d , we substitute the PD control law $\theta = -K_p x - K_d \dot{x}$ back into the state-space model of Section 3.2. If we can get the state-space matrix and its eigenvalues, we can analyze if the closed-loop controller has given us the stability we want. For the x -axis, equations (3.3) and (3.4) become $Dz = A'z$ (read as A_{prime}), which we will then find the eigenvalues for:

$$\dot{x}_1 = x_2 \quad (5.1)$$

$$\dot{x}_2 = (5g/7) \begin{bmatrix} -K_p * x_1 \\ -K_d * x_2 \end{bmatrix} = -\omega_n^2 x_1 - 2\zeta\omega_n x_2 \quad (5.2)$$

In matrix form, the closed-loop system is $Dz = A'z$ with:

$$A' = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix}$$

$$\det(A' - \lambda I) = \lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2 = 0$$

$$\begin{vmatrix} \lambda & 1 \\ -\omega_n^2 & -2\zeta\omega_n - \lambda \end{vmatrix}$$

$$\lambda = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} \quad (5.5)$$

For any $\zeta > 0$ and $\omega_n > 0$, both eigenvalues have strictly negative real part: $\text{Re}(\lambda) = -\zeta\omega_n < 0$. By Theorem 3.4.1, since all eigenvalues of A_{cl} have negative real parts,

the equilibrium $x = 0$ is an attractor. The system is now stable: any ball displacement will now be driven back to the center!

For the simulation in this project, $\zeta = 0.65$ and $\omega_n = 2.5$ rad/s were chosen, giving the underdamped case. By Theorem 3.3.6 since $\alpha = -\zeta\omega_n < 0$, the trajectories in the phase plane spiral toward the origin. The ball returns to center in an oscillatory fashion, overshooting the setpoint before settling. The eigenvalues are $\lambda = -1.625 \pm 1.904i$, which are complex with negative real part, confirming a stable spiral. This matches the behavior observed in the MATLAB phase portrait: the trajectory in the $(x, \dot{x})$ plane traces a clockwise inward spiral converging to the origin.

6. MATLAB Simulation and Results

Four MATLAB scripts were developed to simulate and visualize the system (attached at the end of this paper).

Figure 2 shows the static summary output produced by the MATLAB runner after the animation completed, with parameters $\omega_n = 2.5$ rad/s, $\zeta = 0.65$, and initial conditions $x_0 = 0.25$ m, $y_0 = 0.15$ m. The three panels display ball position versus time, the 2-D ball trajectory on the plate surface colored by normalized time, and the PD controller tilt commands.

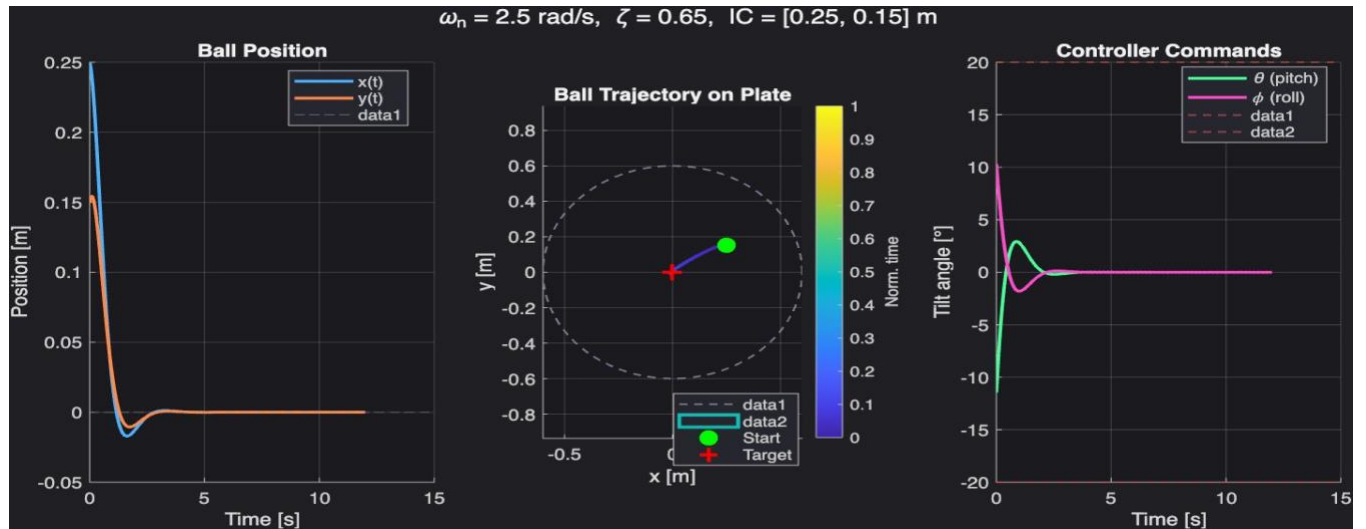


Figure 2. MATLAB simulation summary: ball position time history (left), 2-D plate trajectory coloured by normalised time (centre), and PD controller tilt commands (right). Parameters: $\omega_n = 2.5 \text{ rad/s}$, $\zeta = 0.65$, $IC = [0.25, 0.15] \text{ m}$.

7. Conclusion

Through a complete mathematical analysis of ball stabilization on a Stewart platform, we were able to control a ball on a Stewart platform. Rotation matrices and linear systems (from Chapter 1: Linear Algebra) were used to describe the platform orientation and compute leg lengths. The Laplace transform tools of Chapter 2 converted the ball dynamics into a transfer function and enabled systematic PD controller design. Eigenvalue analysis and phase portrait theory from Chapter 3 (Dynamical Systems) confirmed that the closed-loop system is asymptotically stable for all positive gains, and classified the transient response as a stable spiral, stable node, or overdamped node depending on the damping ratio. MATLAB simulations validated all theoretical predictions and provided an animated visualization of the ball returning to center on the tilting platform, and a stable controller was achieved.

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